



MODULE 6 ASSIGNMENT - FINAL PROJECT

ALY6040 - DATA MINING APPLICATIONS

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INTRODUCTION:

Detecting fraudulent credit card transactions represents a pressing concern for banking organizations, as unauthorized charges result in substantial monetary losses and damage to institutional credibility. The rapid growth of electronic payment platforms has generated unprecedented transaction volumes, rendering human review impossible for fraud prevention. Machine learning algorithms offer a powerful solution by recognizing intricate patterns within transaction histories to flag suspicious activities.

This analysis examines a publicly accessible dataset of credit card transactions featuring anonymized variables and fraud classification labels. The main aim is to implement and evaluate several tree-based predictive algorithms including Decision Trees, Random Forests and Gradient Boosting Machines for identifying fraudulent charges. Because legitimate transactions vastly outnumber fraudulent ones in the dataset, the study emphasizes performance measures beyond overall accuracy, incorporating F1-scores, ROC curve analysis, and confusion matrices. The ultimate objective is determining which algorithm performs best for fraud identification and delivering practical insights from the findings.

PROBLEM STATEMENT AND RESEARCH QUESTIONS

Financial institutions face an ongoing challenge in detecting fraudulent credit card transactions, largely because fraudulent activity represents only a very small fraction of all transactions. This extreme class imbalance makes fraud detection particularly difficult, as models can achieve high overall accuracy while still failing to identify the relatively rare but costly fraudulent cases.

Missing these transactions can lead to direct financial losses and can weaken customer trust in an institution's ability to safeguard financial data.

At the same time, fraud detection systems must function efficiently in real-world environments. While identifying fraudulent transactions is essential, incorrectly flagging legitimate transactions can increase investigation costs and negatively affect the customer experience. As a result, financial institutions must carefully balance minimizing missed fraud cases with controlling false positives. To address these challenges, this project focuses on evaluating multiple machine learning models within a highly imbalanced classification setting to understand both their predictive performance and practical trade-offs.

In particular, this analysis is guided by the following research questions:

1. Can machine learning models accurately identify fraudulent credit card transactions within a highly imbalanced dataset while minimizing missed fraud cases?
2. Which machine learning model provides the best balance between detecting fraudulent transactions and avoiding false positives in an imbalanced classification setting?
3. How can the results of fraud prediction models be translated into actionable insights that support real-world decision-making for financial institutions?

Beyond predictive accuracy, the project also emphasizes model interpretability as a critical requirement for deployment in financial contexts. Understanding why a model flags a transaction as fraudulent helps build trust with stakeholders and ensures that analytical results can be translated into meaningful, actionable insights rather than remaining purely technical outputs.

ANALYSIS:

In this project, the primary goal was to build a machine learning model capable of detecting fraudulent credit card transactions. The dataset used for this analysis contains 284,807 transactions, of which only 492 are labeled as fraudulent, highlighting the extreme rarity of fraud

events. This severe class imbalance presents a major challenge, as models can achieve high overall accuracy by simply predicting most transactions as non-fraud while failing to identify fraudulent cases.

To address this issue, the analysis focused on modeling techniques and evaluation metrics that prioritize the identification of fraud cases rather than relying solely on overall accuracy, which can be misleading in highly imbalanced settings. Emphasis was placed on metrics such as recall, precision, and related performance measures that better reflect a model's ability to detect fraudulent transactions while managing false positives. This approach directly supports the project's research questions, which aim to evaluate whether machine learning models can effectively identify fraud in an imbalanced dataset and determine which models provide the best balance between fraud detection and false alarms.

Step 1: Data Review and Cleaning

The analysis began with a review of the dataset to confirm its suitability for machine learning. All variables in the dataset were numerical, and the target variable clearly identified whether a transaction was fraudulent or legitimate. No missing or duplicate values were present, and the data was already well-structured for analysis. Because the dataset did not require additional cleaning or transformation, it was used directly for modeling. This step was important to ensure data quality and to avoid unnecessary preprocessing that could alter the underlying patterns in the data and potentially affect model performance.

Step 2: Exploratory Analysis and EDA Insights

Exploratory data analysis conducted in Module 3 played an important role in guiding the modeling process. The EDA confirmed that the dataset was highly imbalanced, with fraudulent

transactions accounting for approximately 0.17% (less than 0.2%) of all observations, highlighting the limitations of relying on overall accuracy as an evaluation metric. As a result, metrics such as F1-score and ROC-AUC were emphasized to better assess model performance in identifying fraud cases. The EDA also showed that certain variables exhibited stronger patterns associated with fraudulent behavior, suggesting that fraud detection is driven by a small subset of influential features rather than all variables equally. These insights directly informed both model selection and the focus on interpretability used later in the analysis.

Step 3: Train/Test Split

To directly address the project's research questions, the dataset was divided into training and testing sets using a stratified split. Because fraudulent transactions represent only a small fraction of the data, preserving the original proportion of fraud and non-fraud cases in both datasets was essential. A stratified split ensures that rare fraud cases are adequately represented during both model training and evaluation, allowing for more reliable performance assessment. This approach supports meaningful comparison across machine learning models when determining which method best balances fraud detection with the control of false positives. The training set was used to build and tune the models, while the testing set was reserved for evaluating performance on unseen transactions, reflecting the real-world conditions under which financial institutions must make fraud detection decisions.

Step 4: Models Tested

To address the research question related to model effectiveness and comparison, multiple machine learning models were trained and evaluated, including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, LightGBM, and CatBoost. Logistic Regression was used as a baseline model to establish a simple performance benchmark against which more

complex models could be compared. Tree-based and ensemble models were selected because they are well suited for capturing nonlinear relationships and interactions commonly found in transactional data, and they often perform well in imbalanced classification settings. All models were implemented using Python and standard machine learning libraries to ensure consistent training and evaluation. Evaluating multiple modeling approaches made it possible to identify which model provides the best balance between detecting fraudulent transactions and limiting false positives, a key requirement for practical and scalable fraud detection systems.

Step 5: Evaluation metrics

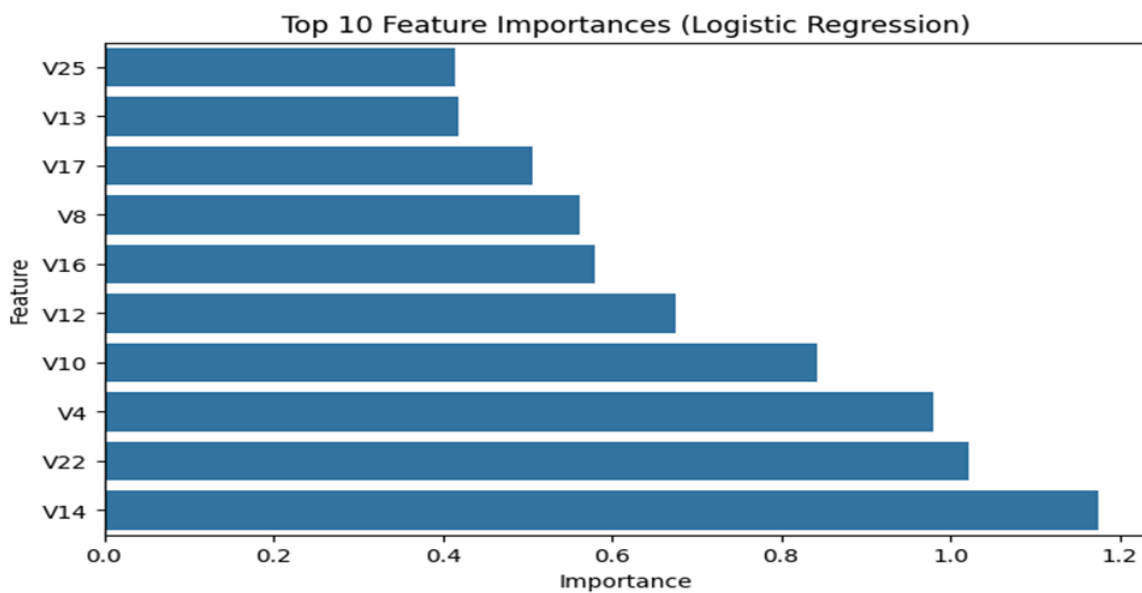
Evaluation metrics were selected carefully to ensure the research questions could be addressed accurately and meaningfully. Given the highly imbalanced nature of the dataset, overall accuracy alone would be misleading, as models could perform well on paper while failing to detect fraudulent transactions. Instead, metrics such as F1-score, ROC-AUC, and confusion matrices were used to evaluate each model's ability to identify fraud while controlling false positives. The F1-score was particularly important because it balances precision and recall, providing insight into how effectively a model detects fraud without generating excessive false alarms. ROC-AUC offered a broader view of model performance by measuring the ability to distinguish between fraudulent and non-fraudulent transactions across different decision thresholds. Together, these metrics enabled a fair comparison of models and supported the interpretation of results in a way that is relevant for real-world decision-making by financial institutions.

RESULTS (MODEL PERFORMANCE AND KEY FINDINGS)

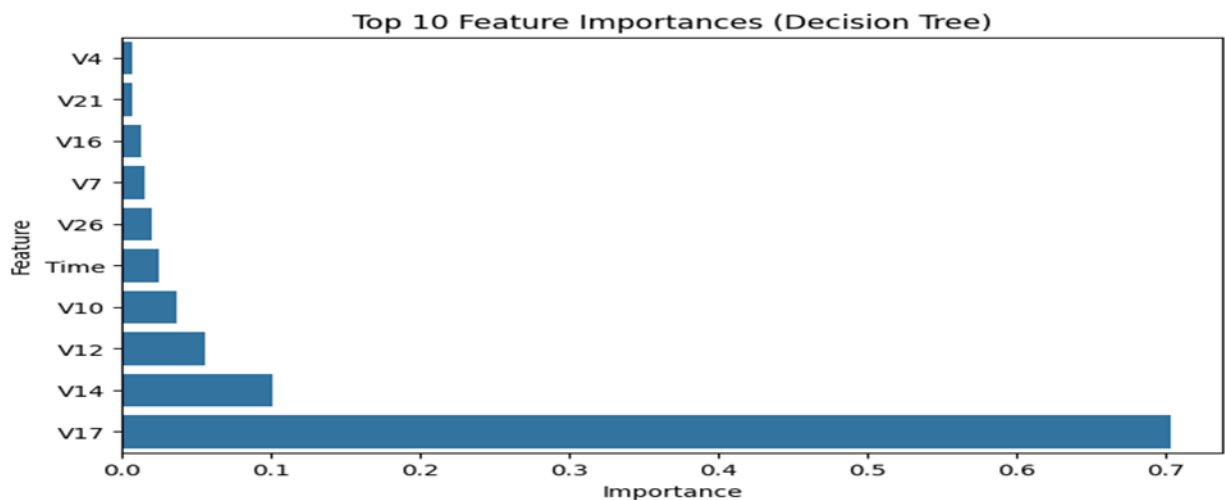
This section presents the performance results of all evaluated machine learning models and connects the findings directly to the research questions through quantitative metrics,

visualizations, and feature importance analysis. The results focus on assessing how effectively machine learning models detect fraudulent credit card transactions in a highly imbalanced dataset, identifying the model that best balances fraud detection and false positives, and demonstrating how model outputs can be interpreted to support real-world decision-making.

To address the first research question, the results show that machine learning models vary significantly in their ability to detect fraudulent transactions despite achieving similarly high overall accuracy. Logistic Regression, used as a baseline model, achieved high accuracy of approximately 97.5 percent but produced a very low F1-score of around 0.11. This indicates that while the model correctly classified most transactions overall, it failed to identify fraudulent transactions effectively. Despite a relatively strong ROC-AUC value of approximately 0.97, the low F1-score suggests that the model made very few fraud predictions under the default classification threshold, resulting in poor recall. These findings demonstrate that not all machine learning models are suitable for fraud detection in highly imbalanced datasets.

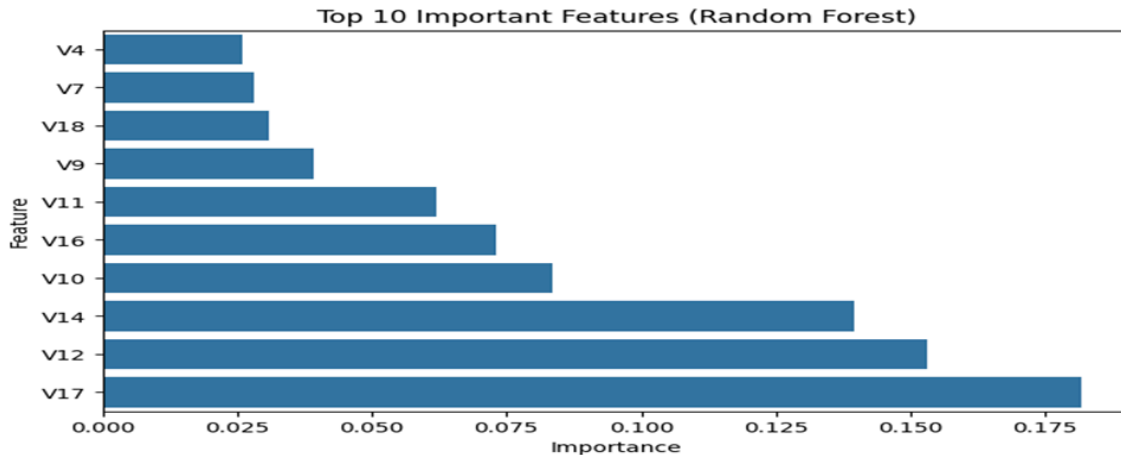


The Decision Tree model showed a substantial improvement in fraud detection performance compared to Logistic Regression. It achieved very high accuracy of approximately 99.94 percent and a strong F1-score of around 0.82, indicating a better balance between detecting fraud and avoiding false positives. The model also had minimal runtime, making it computationally efficient. Feature importance analysis revealed that V17 played a dominant role in the decision-making process, with additional contributions from V14, V12, and V10. These results align with earlier exploratory data analysis findings, which indicated that fraud detection relies heavily on a small number of influential variables. However, because Decision Trees are single-model learners, they are more prone to overfitting and may be sensitive to changes in the data.



In response to the second research question, ensemble-based models demonstrated the strongest overall performance. The Random Forest model delivered the best results among all evaluated models, achieving near-perfect accuracy of approximately 99.95 percent and the highest F1-score of around 0.84. This indicates superior fraud detection capability while maintaining control over false positives. Although the runtime was longer than that of a single Decision Tree, the improvement in predictive performance justifies the additional computational cost. Feature

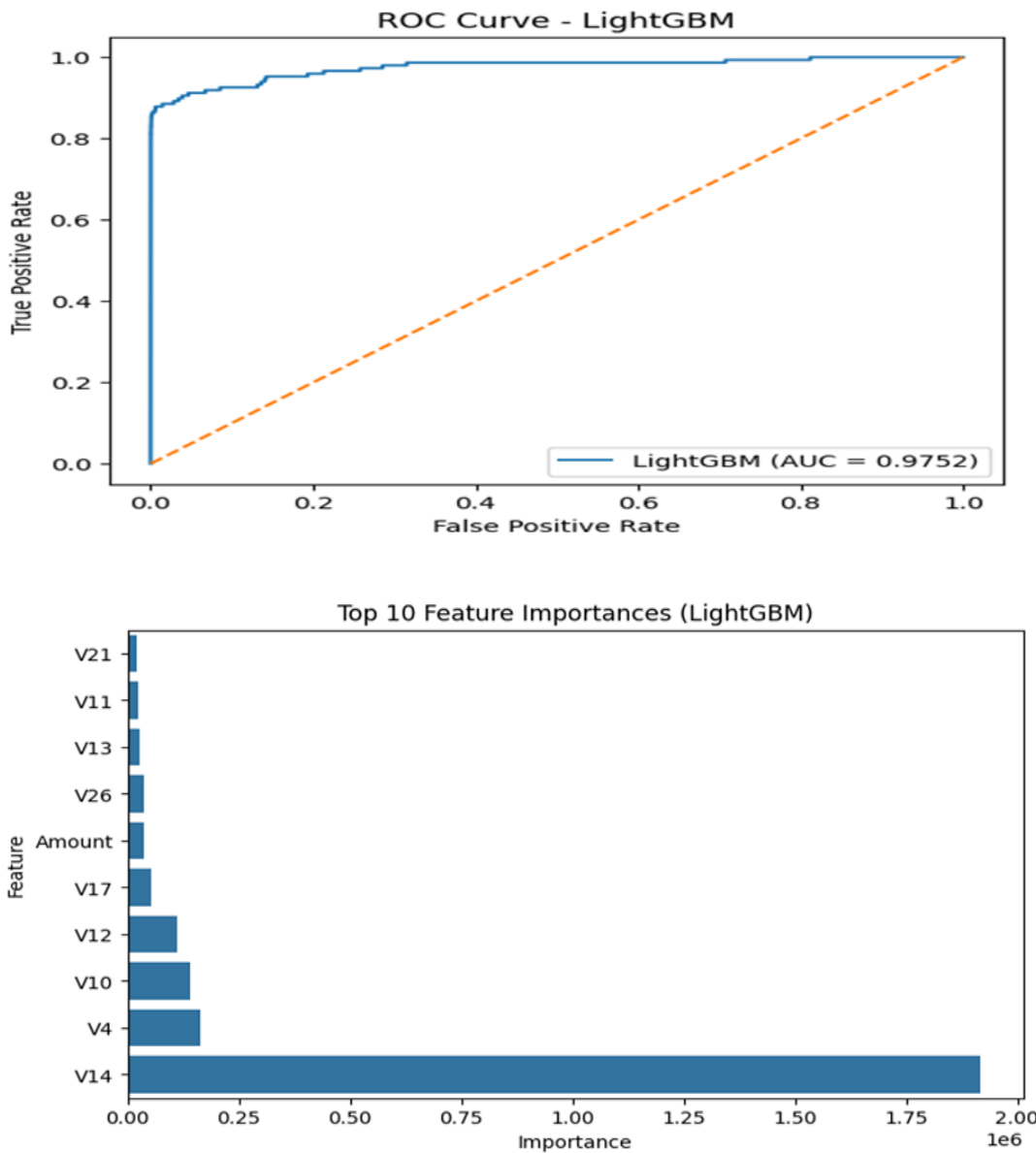
importance analysis consistently identified V17, V12, V14, and V10 as the most influential variables. The ensemble structure of Random Forest enables it to capture complex patterns while reducing overfitting by averaging across multiple trees.



The Gradient Boosting model achieved very high accuracy of approximately 99.84 percent but performed poorly in terms of fraud detection, with a low F1-score of around 0.27 and significantly longer runtime. These results indicate that while the model effectively learned dominant non-fraud patterns, it struggled to correctly identify fraudulent transactions under default settings. This suggests that Gradient Boosting is highly sensitive to extreme class imbalance and requires additional tuning, such as class weighting or threshold adjustment, to be effective in fraud detection scenarios.

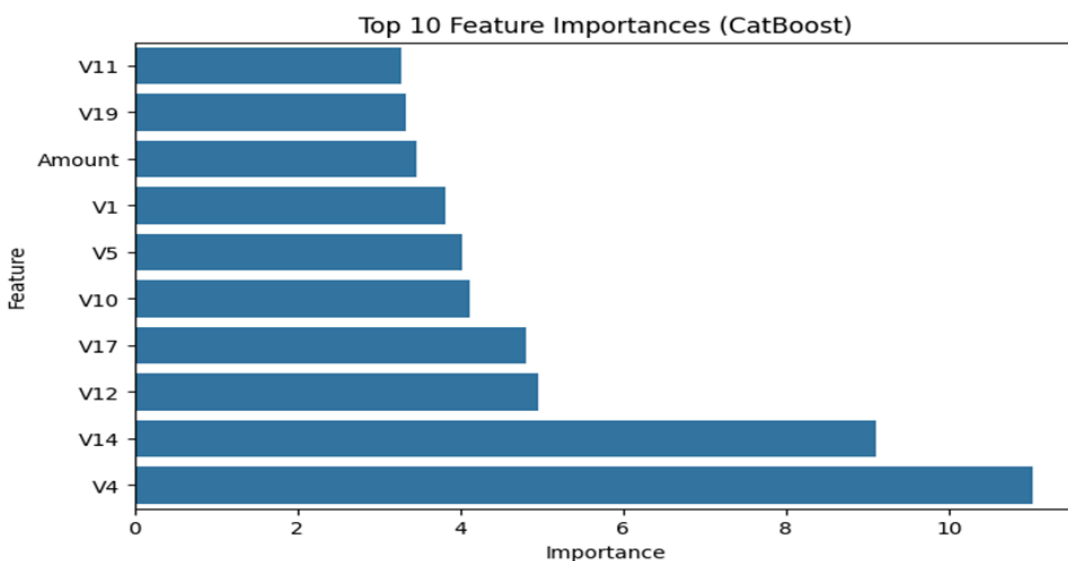
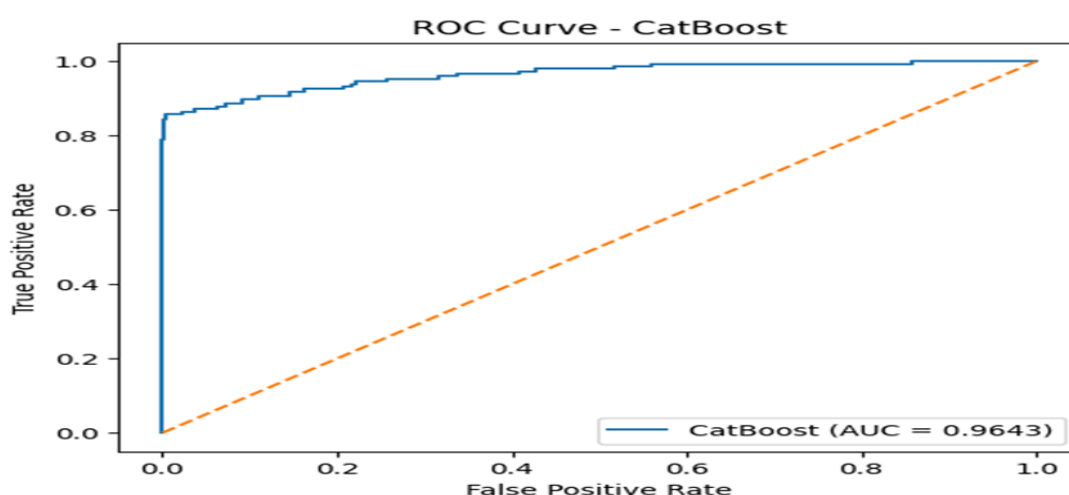
LightGBM demonstrated excellent overall performance and provided strong evidence in support of all three research questions. The model achieved a high F1-score of approximately 0.85 and the highest ROC-AUC value of about 0.98, indicating excellent discrimination between fraudulent and non-fraudulent transactions. Runtime was relatively low compared to other ensemble methods, making LightGBM suitable for large-scale or near-real-time fraud detection systems. Feature importance analysis showed that V14 was the dominant predictor, followed by

V4, V10, and V12. The presence of the Amount variable among important features highlights the role of transaction value in identifying suspicious behavior. These findings closely align with patterns identified during exploratory data analysis and demonstrate how model outputs can be translated into meaningful insights.



CatBoost also performed strongly, achieving very high accuracy of approximately 99.94 percent, a solid F1-score of around 0.81, and a ROC-AUC value of approximately 0.96. The model

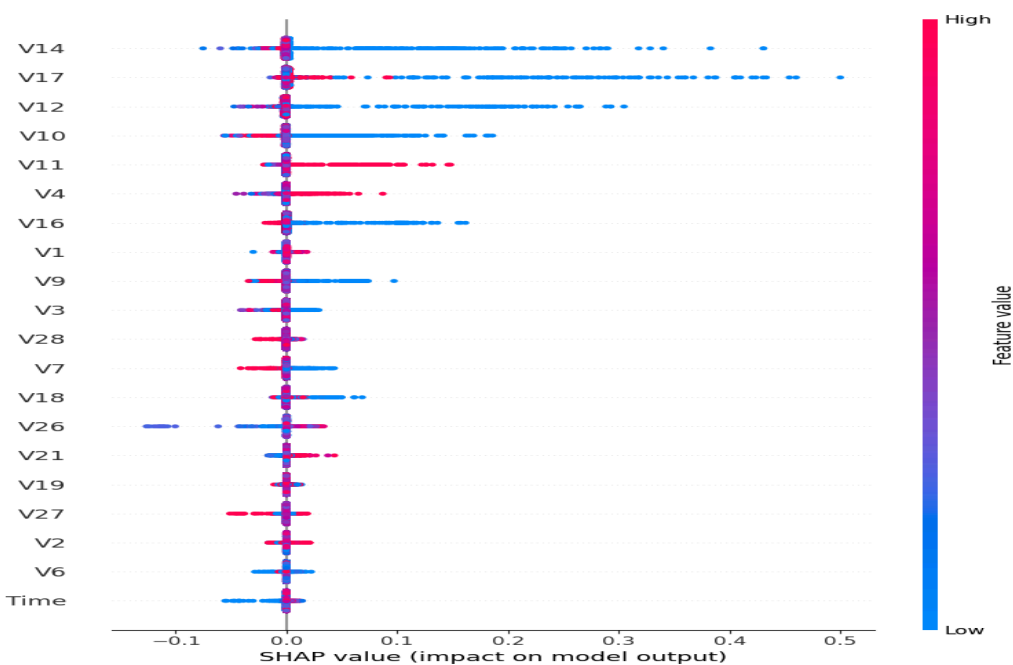
handled class imbalance effectively through the use of class weights and delivered stable performance with moderate runtime. Feature importance results showed that V4, V14, V12, and V17 were among the most influential variables, along with Amount, reinforcing patterns observed across other ensemble models. While CatBoost's performance was slightly lower than Random Forest and LightGBM, its consistency and robustness make it a viable alternative for fraud detection tasks.



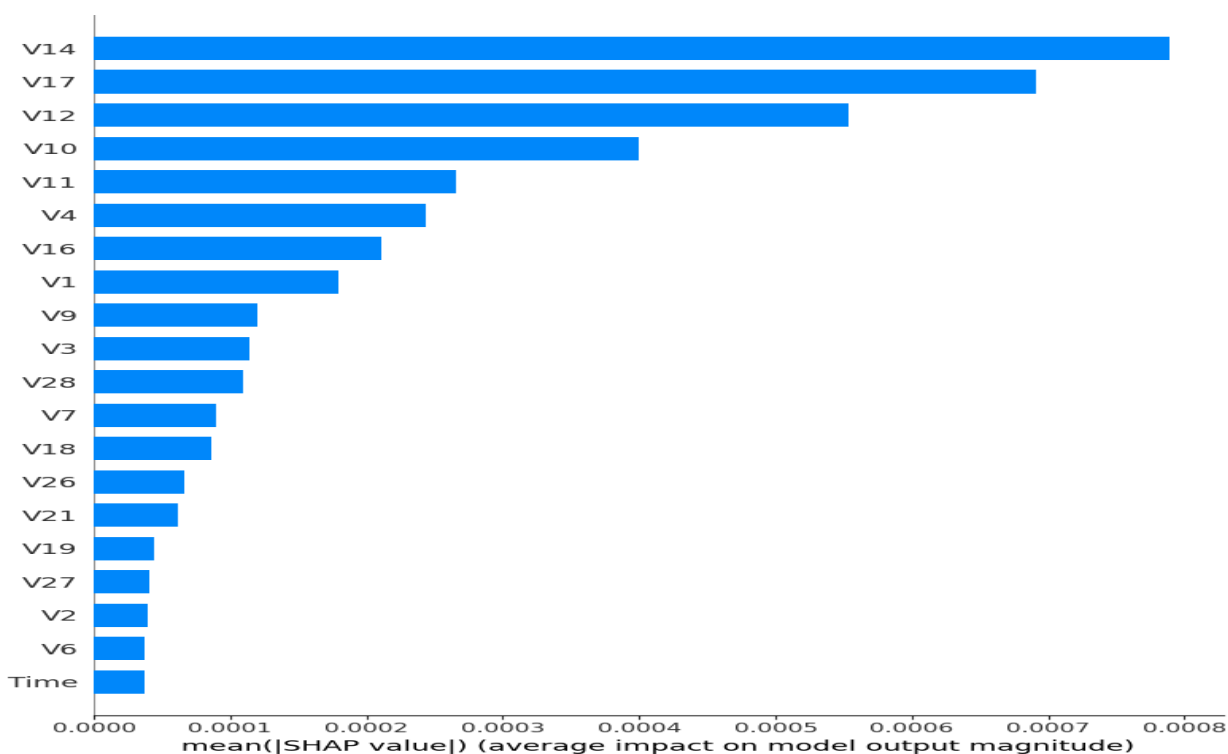
Overall, the results confirm that ensemble machine learning models, particularly Random Forest and LightGBM, are best suited for detecting fraudulent credit card transactions in highly imbalanced datasets. These models not only achieve superior fraud detection performance but also provide interpretable feature importance outputs that align with earlier exploratory insights. By combining strong predictive accuracy with explainability, the models effectively support real-world decision-making for financial institutions, directly addressing all three research questions.

SHAP ANALYSIS - MODEL INTERPRETATION

To better understand what drives the predictive models' decisions, SHAP analysis was carried out to uncover how each feature shapes fraud predictions. Two types of SHAP visualizations were generated for this purpose: a beeswarm summary plot and a bar chart displaying mean absolute SHAP values. When examined side by side, these visuals offer both granular, instance-level insights and a broader, overall picture of how the model arrives at its predictions.



The SHAP summary plot reveals how SHAP values for each feature are distributed across every observation in the dataset. Each dot on the plot corresponds to a single transaction, and its horizontal position reflects both the strength and direction of that feature's impact on the model's output. When a SHAP value falls on the positive side, it nudges the prediction closer to the fraud category; conversely, negative values steer the prediction away from fraud. The color gradient encodes the actual feature values, with red representing higher magnitudes and blue representing lower ones. Among the features displayed, V14, V17, V12, and V10 stand out with the broadest range of SHAP values, signaling that these variables carry the most weight in shaping the model's decisions. A closer look at the patterns reveals that smaller values of these features tend to elevate fraud risk, while larger values generally point the model toward a non-fraud classification.



The bar chart of mean absolute SHAP values offers a high-level ranking of feature importance by averaging the magnitude of each feature's SHAP contributions across all transactions. This view reinforces the finding that only a handful of PCA-transformed features are doing the heavy lifting in the model's predictions. V14 and V17 clearly lead the pack as the most impactful variables, with V12 and V10 following closely behind, while features like Time and several lower-ranked components barely register any meaningful contribution. The sharp decline in importance beyond the top few features suggests that the model leans heavily on a tight cluster of transaction behavior signals rather than drawing from broadly scattered patterns.

Overall, the SHAP analysis aligns with and strengthens the conclusions drawn from conventional feature importance techniques, while adding a deeper layer of transparency by exposing not just how much each feature matters but also the direction in which it pushes predictions. These insights bolster trust in the model's decision-making and confirm that it is picking up on genuine, substantive patterns tied to fraudulent activity rather than relying on arbitrary noise.

RECOMMENDATIONS

The findings from this study confirm that machine learning models are well-equipped to detect patterns linked to fraudulent credit card transactions, even when dealing with heavily skewed class distributions. Across the board, ensemble tree-based approaches consistently delivered superior results compared to simpler alternatives, particularly when evaluated through metrics like F1-score and ROC-AUC that better reflect real-world detection performance. Although accuracy figures appeared impressive for every model tested, this measure turned out to be deceptive given the overwhelming proportion of legitimate transactions in the data. Metrics

designed to spotlight how well models handle the minority class painted a far more honest picture of each model's fraud detection ability.

Of all the methods examined, sophisticated ensemble techniques namely Random Forest, LightGBM, and CatBoost demonstrated the sharpest ability to distinguish between fraudulent and legitimate transactions. Interpretability analysis using SHAP further uncovered that a narrow group of PCA-derived features, most notably V14, V17, V12, and V10, repeatedly played the largest role in driving fraud predictions. This points to the conclusion that fraudulent behavior follows recognizable transaction patterns that tree-based models are particularly adept at capturing. Beyond ranking feature importance, SHAP also shed light on how these features operate directionally, revealing that lower values in certain variables substantially raise the probability of fraud.

From a real-world application perspective, these results carry an important message for financial institutions: evaluation frameworks for fraud detection systems should center on recall and F1-score rather than leaning on accuracy alone. Failing to flag a fraudulent transaction can translate into significant monetary losses, while false alarms, though inconvenient, are typically manageable through follow-up verification steps. With this in mind, decision thresholds should be thoughtfully adjusted to strike a workable balance between operational overhead and effective fraud prevention.

Organizations looking to strengthen their fraud detection capabilities would benefit from adopting ensemble-based models like LightGBM or CatBoost as core components of a larger detection strategy. Pairing these models with explainability tools such as SHAP can go a long way in promoting transparency, earning stakeholder confidence, and meeting regulatory

expectations. Looking ahead, incorporating richer contextual data — such as merchant type, transaction frequency, geographic indicators, or a customer's spending history — could push detection accuracy even further. Routine model monitoring and periodic retraining should also be standard practice to keep pace with the constantly shifting tactics used by fraudsters.

CONCLUSION

This study put several machine learning techniques to the test on a heavily imbalanced credit card fraud dataset to gauge how well each one can spot fraudulent transactions. Logistic Regression was used as a baseline for comparison, while Decision Trees, Random Forests, Gradient Boosting, LightGBM, and CatBoost were brought in to tackle progressively more complex patterns hidden in the data. The results underscore a critical takeaway for imbalanced classification tasks: strong accuracy numbers alone do not necessarily translate to effective fraud detection.

Ensemble tree-based models proved to be the clear frontrunners, outperforming simpler methods especially when measured through F1-score and ROC-AUC. Both feature importance rankings and SHAP analysis confirmed that a compact group of transaction-related features consistently steered model predictions, which indicates that these models are latching onto authentic behavioral signals rather than fitting to random noise. Incorporating explainability techniques added a valuable layer of interpretability, making it easier to understand and trust how each model arrives at its conclusions.

Altogether, this analysis shows that machine learning-driven fraud detection systems can deliver considerable value when they are rigorously evaluated, properly fine-tuned, and clearly interpreted. By pairing powerful ensemble models with explainability tools and performance

metrics that reflect actual business priorities, organizations can sharpen their ability to catch fraudulent transactions without sacrificing transparency or operational efficiency. Moving forward, exploring supplementary data sources, experimenting with different approaches to handling class imbalance, and investigating real-time deployment strategies would all be worthwhile avenues for pushing fraud detection capabilities further.

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